TWWIOS

DEVELOPER HANDBOOK

Relay

WhatsApp campaigns from one Excel sheet. · **Status:** Most complete. Main gap: real Meta testing.

Read first. The code in the Relay zip is written and the server starts, but it has **not been tested against the real external services** listed below. Build the missing parts, connect each service with a test account, then test with real data before any customer uses it.

Languages (built in): Templates can be sent in each contact's language: add a lang column and set perContactLang on the campaign (needs an approved template in each language).

1. What is in the zip

- `code/api/server.js` standalone Relay server (Node.js, Express, MongoDB)
- `code/api/pulse/auth.js`
- `code/api/pulse/billing.js`
- `code/api/pulse/events.js`
- `code/api/pulse/hooks.js`
- `code/api/pulse/mail.js`
- `code/api/pulse/routes.js`
- `code/api/pulse/sender.js`
- `code/api/pulse/shield.js`
- `code/api/pulse/social.js`
- `code/api/pulse/store.js`
- `code/api/pulse/wa.js`
- `code/web/` Relay screens for the Next.js app

Run: `cd code/api` → `cp .env.example .env` → `npm install` → `npm start` → open `http://localhost:4000/health`. Without MongoDB it runs in memory for testing.

2. What to build in Relay

Connect a real WhatsApp number

Before launch

`/api/pulse/wa/embedded, /wa/connect · pulse/wa.js`

- Create a Meta app with WhatsApp product; set FB_APP_ID, FB_APP_SECRET, WA_CONFIG_ID
- Finish Business verification and Tech Provider review (scripts in docs/META_APPROVAL.md)
- Run Embedded Signup with a real number; save WABA id and token (encrypted with ENC_KEY)

Done when: A customer connects their own number from the app.

Webhooks with real traffic

Before launch

`/api/pulse/webhook · pulse/events.js`

- Set webhook URL and WA_VERIFY_TOKEN in Meta
- Check signature with WA_APP_SECRET
- Confirm sent, delivered, read and reply events update the campaign

Done when: Campaign screen shows live delivered/read/reply numbers.

Real campaign test

Before launch

`pulse/sender.js, pulse/shield.js`

- Own number + 20 contacts who agreed
- Check rate limit, first small wave (canary), opt-out keyword STOP

Done when: All 20 receive; STOP removes the contact.

Template status in the app

Before launch

`pulse/wa.js listTemplates · WaApp.jsx`

- Show PENDING / APPROVED / REJECTED and the rejection reason
- Refresh status from Meta

Done when: User sees why a template was rejected.

Usage-based charging

After launch

`pulse/billing.js, pulse/shield.js`

- Count conversations by category (marketing, utility)
- Show usage vs plan; block or bill overage

Done when: Usage page matches Meta's conversation count.

Instagram / Facebook comment-to-DM live

After launch

`pulse/social.js`

- Get Meta permissions reviewed: instagram_manage_messages, pages_messaging, instagram_manage_comments
- Test on a real page and account

Done when: A keyword comment gets a private message reply.

Store and CRM connectors

After launch

`new api/connectors/`

- Shopify / WooCommerce: order update messages
- HubSpot / Zoho: import contacts

Done when: Order placed in Shopify triggers a WhatsApp update.

3. Shared platform work (do once for all products)

One customer app with agent menu

Before launch

`web/app/(app)/ · design canvas "TWWIOS App"`

- Build the app shell from the design: left menu with Home, the 5 agents, Plan & billing
- Read the company plan from /api/pulse/me and show only paid agents; others show "Add to plan"
- "Add to plan" opens checkout and unlocks the agent after payment webhook

Done when: A customer signs up once, buys Relay, then adds Mira without a new login.

Website pages in Next.js

Before launch

`design/website-final.html → web/app/(site)/`

- Copy sections (hero, platform, transformation, solutions, how it works, FAQ) into React components
- Keep the same CSS tokens and fonts
- Point "Try" and "Get started" buttons to the real app

Done when: Website matches the preview link and every button works.

Billing: one subscription, add-ons, USD

Before launch

`api/pulse/billing.js`

- Add Stripe next to Razorpay (Stripe for USD, Razorpay for INR)
- One subscription per company with one line item per agent
- Webhooks update the plan; failed payment downgrades after a grace period

Done when: Adding or removing an agent changes the next invoice correctly.

Team members and roles

After launch

`api/pulse/routes.js, auth.js`

- Invite by email; roles owner, admin, member
- Members can be limited to certain agents

Done when: An invited member logs in and sees only allowed agents.

Production setup

Before launch

`DEPLOY.md, scripts/backup.sh`

- MongoDB Atlas with daily backups
- Secrets in the host, never in git; HTTPS; health checks
- Error tracking (Sentry) and uptime alerts

Done when: App runs with NODE_ENV=production and passes the start-up checks.

Tests in CI

Before launch

`api/*test.mjs`

- Run existing tests on every push (GitHub Actions)
- Add tests for Settle, Scribe and Vox (none today)

Done when: All tests pass on every push.

Legal and company details

Before launch

`web/lib/company-details.js`

- Terms, Privacy Policy, Data Processing Agreement, data-deletion page
- Company address, email, phone, GSTIN

Done when: Pages are live and linked in the footer.

4. External services and APIs to sign up for

Not included in the code. Open these accounts and put their keys in `code/api/.env`.

SERVICE / API	USED FOR	WHERE

Meta WhatsApp Business Platform (Cloud API)	Sends messages, receives replies and read receipts	`developers.facebook.com`
Meta Tech Provider approval + Business verification	Lets customers connect their own WhatsApp numbers	`business.facebook.com`
Instagram / Facebook Messaging API permissions	Comment-to-DM	`developers.facebook.com`
Shopify / WooCommerce API (later)	Order updates	`shopify.dev / woocommerce.com`
HubSpot / Zoho CRM API (later)	Contact sync	`developers.hubspot.com / zoho.com/crm`
MongoDB Atlas	Database	`mongodb.com/atlas`
Hosting server	Runs the app (Render, Railway or a VPS)	`render.com / railway.app`
Stripe	USD subscriptions	`stripe.com`
Razorpay	INR subscriptions and payment links	`razorpay.com`
Resend	Emails: sign-up, password reset, alerts	`resend.com`
Sentry (optional)	Error alerts	`sentry.io`

5. Environment keys

- `MONGODB_URI`
- `AUTH_SECRET (32+ random chars)`
- `ENC_KEY (32+ random chars)`
- `PUBLIC_URL`
- `CORS_ORIGIN`
- `FB_APP_ID`
- `FB_APP_SECRET`
- `WA_CONFIG_ID`
- `WA_VERIFY_TOKEN`
- `WA_APP_SECRET`
- `RAZORPAY_KEY_ID / RAZORPAY_KEY_SECRET / RAZORPAY_WEBHOOK_SECRET`
- `RAZORPAY_PLAN_STARTER / RAZORPAY_PLAN_GROWTH`
- `RESEND_API_KEY, MAIL_FROM`
- `PULSE_ADMIN_KEY`

6. Final checklist before going live

- All "Before launch" items above are done
- Every external service tested with a real account
- Tested end to end with your own data, then one friendly customer
- Production keys set, backups on, error alerts on
- Terms, Privacy Policy and company details live on the website